
Integrable Systems — Exercise Sheet 5

The Liouville–Arnold theorem

Variant: Questions only

Conventions, as in Lectures 2–5: $\{f, g\} = \sum_i (\partial_{q_i} f \partial_{p_i} g - \partial_{p_i} f \partial_{q_i} g)$, $\omega = \sum_i dq_i \wedge dp_i$. The theorem's hypotheses are named as in the lecture: $F_1, \dots, F_n$ in involution, Σ a connected component of a level set, (I) the dF_i independent at every point of Σ , (C) Σ compact.

Exercise 1 (independence and involution). (i) Let F be any smooth function on a phase space and g any smooth function of one variable. Show that $\{F, g(F)\} = 0$ and that dF and $d(g(F))$ are linearly dependent at every point. What does this say about counting functions in involution? (ii) On $T^*\mathbb{R}^2$ with $q \neq 0$, let $H = \frac{1}{2}|p|^2 + V(|q|)$ with $V'(r) > 0$ for $r > 0$, and $\ell = q_1 p_2 - q_2 p_1$. Show that $\{H, \ell\} = 0$, and that dH and $d\ell$ are linearly dependent exactly at the points where p is perpendicular to q with $|p|^2 = |q| V'(|q|)$: the circular orbits. (iii) Take the Kepler potential $V = -1/r$. For fixed $\ell \neq 0$, show that the smallest energy is $-1/(2\ell^2)$, reached on the circular orbit of radius ℓ^2 , and that the level set $\{H = -1/(2\ell^2), \ell = \ell\}$ is exactly that circle. Which hypothesis of the theorem fails there, and which conclusion?

Exercise 2 (deleting involution). On $\mathbb{R}^4$ let $F_1 = \frac{1}{2}(q_1^2 + q_2^2 + p_1^2 + p_2^2)$ and $F_2 = q_1$, and fix $c > 0$ and a with $a^2 < 2c$. (i) Show that F_1, F_2 are not in involution. (ii) Show that the level set $\Sigma = \{F_1 = c, F_2 = a\}$ is a 2-sphere, compact and connected, and that dF_1, dF_2 are independent at every point of it. (iii) Show that Σ is not Lagrangian and not invariant under the flow of F_1 . Which conclusion of the theorem fails, and why does this not contradict it?

Exercise 3 (one degree of freedom, components, and a cylinder). (i) (Stage (iv) with $n = 1$.) Let Σ be a compact connected component of a level set of H on which $dH \neq 0$, and $x_0 \in \Sigma$. The flow of H acts on Σ , and the lecture's stage (iii) shows that the orbit of x_0 is all of Σ . Show that the stabiliser $\Lambda = \{t : \varphi_t(x_0) = x_0\}$ is either $\{0\}$ or $\tau\mathbb{Z}$ for some $\tau > 0$, and that compactness excludes $\{0\}$. What is Σ , and what is τ ? (ii) For $H = \frac{1}{2}p^2 + \frac{1}{4}q^4 - \frac{1}{2}q^2$ at $E = -\frac{1}{8}$ (Sheet 1, Exercise 4), how many components has the level set? What does the theorem say about the level set, and what about each component? (iii) The pendulum $H = \frac{1}{2}p^2 - \cos \theta$ at $E > 1$, with θ taken in $\mathbb{R}$ (not modulo 2π): what is each component of the level set, and what is its stabiliser? Which hypothesis fails? What changes on the cylinder? (iv) On $\mathbb{R}^4$ let $F_1 = \frac{1}{2}p_1^2$ and $F_2 = \frac{1}{2}(q_2^2 + p_2^2)$. Show that they are in involution, and describe a component Σ of $\{F_1 = e_1, F_2 = e_2\}$ for $e_1, e_2 > 0$. Find the stabiliser of the action of $\mathbb{R}^2$ on it, and check the lecture's $\Sigma \cong T^k \times \mathbb{R}^{n-k}$.

Exercise 4 (the two-dimensional oscillator, stages (ii)–(iv)). Let $H_i = \frac{1}{2}(p_i^2 + \nu_i^2 q_i^2)$ with $\nu_i > 0$, $i = 1, 2$, and a point x_0 with $H_1(x_0), H_2(x_0) > 0$. (i) Write the action of $\mathbb{R}^2$, $\Phi_{(t_1, t_2)}(x_0) = \varphi_{t_1}^{H_1} \circ \varphi_{t_2}^{H_2}(x_0)$, explicitly. Why does the order of the two flows not matter? (ii) Find the stabiliser Λ of x_0 , and deduce that the level set through x_0 is a 2-torus. (iii) Take the basis $e_1 = (2\pi/\nu_1, 0)$, $e_2 = (0, 2\pi/\nu_2)$ of Λ . Find the angles $\varphi = 2\pi A^{-1}t$ and the two basis cycles, and show that the flow of $H = H_1 + H_2$ is $\varphi(t) = \varphi(0) + (\nu_1, \nu_2)t$.

Exercise 5 ★. *The spherical pendulum's energy–momentum map.* Away from the poles, with the conventions of optional lecture A (unit mass, length and gravity, θ measured from the

upward vertical, so that the height is $\cos \theta$),

$$H = \frac{1}{2} \left(p_\theta^2 + \frac{p_\phi^2}{\sin^2 \theta} \right) + \cos \theta, \quad J = p_\phi, \quad \theta \in (0, \pi).$$

- (i) Show that $\{H, J\} = 0$. With $V_J(\theta) = J^2/(2 \sin^2 \theta) + \cos \theta$, show that for $J \neq 0$ the function V_J has exactly one critical point, a minimum at some $\theta_J \in (\pi/2, \pi)$ with $J^2 \cos \theta_J = -\sin^4 \theta_J$. (ii) Deduce that for $J \neq 0$ the energies reached are exactly $E \geq E_{\min}(J) = V_J(\theta_J)$, and show that $E_{\min}(J) = -1 + |J| + O(J^2)$. (iii) For $J \neq 0$, show that dH and dJ are dependent exactly on the circles $p_\theta = 0$, $\theta = \theta_J$ (the conical motions), and that for $E > E_{\min}(J)$ the level set $\{H = E, J = J\}$ is a 2-torus. (iv) At $J = 0$ the pendulum swings in a vertical plane, and its two equilibria — hanging down and balanced upright — lie on the poles, outside these coordinates. Their energies are -1 and 1 . Show that the value $(J, E) = (0, 1)$ lies inside the image of the map (J, H) , above the curve $E = E_{\min}(J)$.